AMITY INSTITUTE OF INFORMATION TECHNOLOGY

WEEKLY PROGRESS REPORT
FOR THE WEEK COMMENCING (29/06/2026-05/07/2026)

WPR No: 07
Program: MCA

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ENROLLMENT NO.- A010145025215

SEMESTER: 3rd
FACULTY GUIDE NAME: Dr. Garima Bohra

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

TARGETS FOR THE WEEK:

1. Integrate the real-time gamma radiation detector with the live webcam visualization system.
2. Develop a dynamic radiation intensity overlay using a scientific color gradient **(Blue → Green → Yellow → Red)**.
3. Synchronize continuous gamma detector telemetry with live webcam frame rendering for low-latency monitoring.
4. Optimize the digital twin visualization pipeline for stable real-time performance and reduced rendering delays.
5. Establish the foundation for transitioning from a 2D radiation monitoring interface to a 3D digital twin environment.

ACHIEVEMENTS OF THE WEEK:

1. Successfully integrated the gamma radiation detector with the live webcam to create a real-time **“3D digital twin”** visualization.
2. Implemented dynamic radiation intensity overlays based on continuous gamma count-rate telemetry.
3. Developed a real-time color-mapping algorithm for intuitive visualization of radiation intensity levels.
4. Established continuous synchronization between sensor telemetry and live video streaming for environmental radiation monitoring.
5. Identified and analyzed synchronization bottlenecks affecting overlay stability, providing a roadmap for performance optimization.

FUTURE WORK PLANS:

1. Implement buffered sensor-data averaging to reduce measurement noise and stabilize radiation visualization.
2. Optimize frame synchronization and rendering pipelines to eliminate flickering and minimize latency.
3. Develop smooth multi-stage color interpolation for accurate real-time radiation intensity transitions.
4. Extend the current 2D monitoring system into a fully interactive **3D Digital Twin** with spatial radiation mapping.
5. Integrate advanced analytics, volumetric visualization, and real-time environmental radiation localization for intelligent monitoring and decision support.